

CONSTELLATION SOFTWARE INC CSU.

February 13, 2026 - 10:49am EST
by

MarAzul

			2026	2027
Price:	2,325.00	EPS	165	0
Shares Out. (in M):	21	P/E	14	0
Market Cap (in \$M):	36,500	P/FCF	0	0
Net Debt (in \$M):	0	EBIT	0	0
TEV (in \$M):	0	TEV/EBIT	0	0

Description

Good ideas are few and far between. For months we search everywhere with little hope, and then suddenly something weird happens! I try to post stuff when something is obvious, that is the reason why I have lost my VIC status a few times and explains the “reactivation” post (sorry for the trouble, but would appreciate the vote). I think Constellation Software is one of those opportunities; fish in the barrel after you drained the water type stuff.

The Company is well known as it has compounded value at +20% CAGR for years; company went public at \$17/share, a 130x over 20 years. CSU is the vehicle created by Mark Leonard to allocate capital in the vertical software market; they have done +1,000 acquisitions in 30 years.

Over this period the company has used almost no debt; Holdco debt is \$2.5bn while cash is more than that. Also there has been basically no share issuance, with S/O flat at 21.2mm since 2006. As a side note, the company has no SBC and executives own large chunks of stock as they are required to reinvest part of their compensation. The point is that while CSU has been a fast-growing conglomerate, they have not used financial engineering as the source of growth. The company probably could have stayed private; they went public in 2006 because they had taken some private equity money and needed to find an exit to the group.

So what do they actually do? Think of Constellation as a Berkshire type holding company, without the insurance float. They hold thousands of subsidiaries that each operate independently, sharing some resources and knowledge, but besides that they are autonomous.

The company has built a constellation of subsidiaries by acquiring small assets, at low valuations given the size; typically, they pay close to 1-1.5x sales. The company keeps operating as it is, but managers usually receive feedback and comments on best practices. Besides that, I have heard they usually raise prices a bit after the acquisition as they have noticed customers rarely churn; I understand this is not ideal.

The subsidiaries are in the software market, providing solutions for specific niches. For example, I work as an asset manager, and we employ software that divides which assets belong to each account, directs payments, sends notifications, registers all the transactions, generates account statements, sends regulatory data, compliance, etc. There are unique features needed for my industry and country in which we operate, this is not a common software like salesforce that you can use like everybody else. It is almost custom built for our needs. CSU is full of small companies that provide these types of solutions. They often mention the software that runs a country club or a bowling alley. Point is most of their companies offer a niche, custom, specific service for customers they have had relationships for years and are part of the business process; instead of offering a horizontal service that can be used by any company.

Leonard has mentioned that “vertical market software is the distillation of a conversation between a vendor and the customer that has gone on frequently for decades”. This makes total sense, going back to the example, in my company we talk with the software provider often to add features, include new users, solve problems, etc. The solution has a component of technology, but it is dependent on service. The distinction is important because well known software companies like Microsoft, Salesforce, Adobe, etc. have offerings that can be scaled, and used by the masses. In contrast, CSU has focused on niches where the solution is close to custom made, critical, cannot be replicated in every market, etc. One downside is there is little room for huge growth and large TAM, but it is probably more defensible. There is less competition in niches, and also once you have become the custom solution for a critical process the likelihood of losing a client is lower. So we might expect less organic growth and probably more stickiness.

Why now? This conversation about the product is highly relevant because a possible explanation for the 60% decline in share price is AI. Market fears that as AI advances, products like Claude Code might make it easy to build and replace software services offered by CSU’s subsidiaries.

Some think that companies might build their own custom software from scratch, and while it is possible I see it as unlikely given that companies usually focus on their main mission and rely on others for services including technology. Also, it is hard to believe a small company would not only build the code to run the business, risking all the problems that come with it, but also provide the maintenance, patches, etc. It is certainly possible but seems unlikely as a big threat long term for companies offering custom, critical software services.

In a recent call the CEO of Jack and Henry mentioned the following: “what AI does in the development of technology and the development of building whether that be a core system or other very complex solutions that we support in this industry, it's not just as simple as doing things faster. It's way more complicated than that. It creates some concerns for maybe some of the other areas where people are doing seat licenses and other stuff, so some of the other larger enterprise-wide solution sets” They of course provide core banking services which is as sticky as it gets, but the thought process makes sense. If you are providing a critical and custom service then that is sticky and more defensible; seems like CSU was built on this premise and they are well protected with difficult to replace services.

Other threat is that with the rise of AI, building software is so much easier. This will increase the supply of software and competitors, pushing down pricing. This seems like a real threat and makes me a bit nervous. I think CSU has the relationship with the final customer and they will have to use AI to improve their offering and lower costs. This might be a challenge and it is difficult to predict the final outcome. My bet here is that value comes from a strong relationship with the client, understanding their needs and employing technology to better serve that customer base. Will have to keep an eye on it.

In a best case scenario, you can also see CSU having to employ less people in software development, while maintaining the client base, and that gets you to higher margins. Who knows? In a recent call, Mark Leonard mentioned that an AI specialist a few years ago predicted that AI would replace radiologist, so people should stop studying radiology. Instead, over the last decade, the number of radiologists has expanded by 17% and most of them are thriving. The prediction missed that AI would increase productivity and improve care. Point is it is very difficult to predict the future and how it will affect different services, companies, etc.

Certainly AI will have a negative impact on some of CSU’s subsidiaries, given they own businesses in various fields. Would think that the majority of the businesses are somewhat insulated from a catastrophic scenario. The reality is we don’t have a complete view of how much is exposed to AI, etc. as the company operates over 1,000 subsidiaries; we have to think based on probabilities.

Another area of concern is succession. In September 2025, Mark Leonard announced that he was stepping down as CEO due to health issues and would remain in the BoD. He is being replaced by Mark Miller a long time executive, he was actually one of the founders of the

first company CSU acquired. While replacing someone like Leonard is no easy task, I think CSU's structure makes it less troublesome. The subsidiaries operate independently, with their own teams that manage the companies and allocate capital. Leonard has built a system that pushes down capital allocation decisions to the subsidiary level, so there is less pressure at headquarters. Also, the culture is part of the special sauce: ethical, care for the small shareholder, skin in the game, attention to detail, valuation rigor, etc. I believe culture will survive Leonard and is ingrained in the executive team. It seems like the system created by Leonard is special, and should outlive him. CSU will also benefit from his involvement at the board level, and my guess is he can still influence on high level strategic decisions and capital allocation.

Valuation? The combination of AI fears and Leonard's resignation has opened an opportunity to buy into a special company at a modest valuation. CSU trades at less than 15x FCF, the lowest valuation since the company gained some size in 2014 or so. Shares have traded close to 25x for most of the last decade, and +30x at times.

Current levels seem attractive for a company that over the last decade, has grown cash flows per share at 20% cagr. While organic growth has been modest at low single digits, growth has come from reinvesting cash flows at high rates through acquisitions. Most of these are small companies that can be acquired at decent prices and the operations are improved over time.

Let's assume that over the next 5 years, they can grow cash flows at a more modest 12% cagr that would leave us with \$4.2bn in FCF by 2030. Once some of the AI and succession doubts dissipate, I think we can reach 20x FCF which would still be modest for such a high quality business. That would yield \$84bn in market cap or a 2.3x.

In case things don't work out as expected, and software is indeed a melting ice cube as the market thinks, then there is a path for downside protection. Constellation is a capital allocation machine, and my bet is they will do the best to allocate cash flows if the businesses are in perpetual decline. That might be paying less for acquisitions in the same industry or allocating cash flows to new endeavors. 10 years ago Leonard even mentioned that one day VMS would enter the decline phase, and that they would have to manage that process. He gave some examples of media empires that did well like Graham Holdings. They could possibly milk the assets for cash flows, which at current levels would be a 6% yield and try to extend the life of the cash flow stream, while diversifying their revenue base. That transition is tricky, but in case we get that scenario, I think we are in good hands and at less than 15x the market is already discounting that possibility.

Another reason why I like CSU at the moment is that the sector as a whole is in the dump; many names are down +70%. I want to put some capital in the sector in case things turn out to be less bad than expected. CSU seems like a good vehicle given they are specialists at allocating capital in software. This might be an environment where they can get opportunities from forced sellers in the PE market, bankruptcies, discounted public equity, etc. Also, there is no dilution and management is ethical and aligned.

I do not hold a position with the issuer such as employment, directorship, or consultancy.

I and/or others I advise hold a material investment in the issuer's securities.

Catalyst

- More acquisitions as bad climate continues.
- Good results continue as businesses show resilience
- Leonard comments at AGM calm investors.
- Buybacks? they have indicated no interest in buying back shares, but might change mind

Messages

Subject Organic Growth
Entry 02/13/2026 10:58 AM
Member mrbing

Thanks for this, certainly a topical name.

Here is what I have been unable to reconcile. The Company's own disclosed organic growth rates are very low single digits, about +2-3%, so when we talk about the company compounding at 15-20%+ that is almost entirely due to its roll up model. Almost all of CSU's free cash flow is allocated towards M&A (plus paying earn outs on past deals). So if you are underwriting the Company to continue growing at ~15%, then shouldn't we treat acquisition spend as capex in the FCF number? What multiple of FCF are we paying with that framework, and is that cheap?

Another way I think would be helpful to frame this is what multiple do software companies with flat to LSD organic growth rates trade at, and what premium or discount is CSU? My experience is that when top-line is flat for these businesses (or god forbid, swings to a LSD decline), the valuation multiple collapses to something like 5-6x EBITDA with a high FCF yield.

Appreciate your thoughts

Subject Funding short?
Entry 02/13/2026 11:17 AM
Member Ray Palmer

I have no dog in this race, but this still trades at a premium valuation versus most other software

If software is dead from AI or permanently impaired, this goes down more than everything else given that premium will bleed out

If the AI fears are overblown, doesn't everything else snap back faster / harder?

Subject Re: Organic Growth
Entry 02/13/2026 11:19 AM
Member MarAzul

Good and fair points. Let's assume they would suddenly stop doing acquisitions. We would have a ~3% growth + 7% FCF yield. That seems fair, given the lack of leverage and quality of management. We can play around with the numbers and maybe apply a haircut, but think it would not be significant. Also, if there are no acquisition efforts going forward, one would expect some cost reductions so maybe a small increase in FCF margins? Historically margins have averaged close to 20%. In a no growth scenario, one can imagine that going to 22-25%? Who knows. As you mentioned, the value generation is coming from the acquisitions and it can be interpreted as growth capex. It would be a negative if they don't find opportunities to reinvest the cash flows. The platform is a valuable asset.

Subject Re: Funding short?
Entry 02/13/2026 11:26 AM
Member MarAzul

Thank you Ray, valid points. I am not a software expert, so don't feel comfortable placing bets on specific names that indeed might be cheaper. I want to find a way to allocate capital in the sector given valuations. CSU for me offers a way to participate in a wide range of businesses, and also they can reinvest the cash flows for me. I feel comfortable with the business diversification and quality/integrity of management. At 14x, seems cheap enough but acknowledge that other things might be cheaper.

Subject Uses of capital
Entry 02/13/2026 12:02 PM
Member mike126

I don't know the answer to "the AI question" but it seems to me that in software, things are more binary than they have looked in a while. Either things are good, or bad. And for a company like CSU, that in effect means that soldiering-on with the M&A-driven model increases the risk profile to the shareholder.

The way that CSU could decrease the risk would be to either stockpile cash or give it back as dividends. Buybacks also increase the risk, but the increase in risk there would be less than with M&A. Would be curious to hear your thoughts. Whenever I have invested in or looked at highly M&A-driven businesses that either got in some sort of trouble or were faced by profound market skepticism, what I wanted these mgmt teams (or boards) to do is to put a stop to the M&A for a while (say a year or two, or even three) so that we (by "we" I mean the collective shareholder and investor community) could gain some comfort in the organic growth (i.e. that the growth won't fall off a cliff and turn negative), so that we could see the FCF generation, and so that we could enjoy some immediate fruits of it (e.g. via dividends). I.e. I wanted the company to simplify and reduce the number of moving parts for a while. But of course, this pretty much never happens. M&A-driven mgmt teams / cultures are wired to keep doing what they know best.

E.g. as an aside, Roper (ROP US) recently pivoted towards buybacks, saying that the hurdle rate for M&A is justifiably higher than the returns they think they could get from buybacks.

Subject Re: Uses of capital
Entry 02/13/2026 12:08 PM
Member MarAzul

Thank you for the comments Mike. In this case, organic growth is quite low. They have averaged low single digits for years. A possible explanation is the quality of the assets is not great. Makes sense given the multiples they pay are quite low and that would not be possible if they were great, growing assets. Another explanation is that at times, they buy declining businesses, but given the acquisition price, the IRR still works. They have mentioned the acquisition of Allscripts as an example of a declining business with good IRRs. Regarding stopping acquisitions for a while, and returning cash flows, that would yield decent but "no-great" returns for shareholders; something like 7% fcf yield + 3% growth = 10% return assuming constant multiple.

Subject Re: Re: Uses of capital
Entry 02/13/2026 02:21 PM
Member Cries

My understanding has always been: they are not buying growth. They are buying high retention; targeting deeply embedded vertical saas that has already peaked in terms of penetration.

I guess the new narrative today is that the retention moat just got a whole lot weaker. Im not sure if thats true, and I still think vertical saas & payments is probably among the more sustainable software assets out there.

Subject Re: Funding short?
Entry 02/14/2026 06:47 PM
Member Rtg123

Out of curiosity, what software peers are you referring to that trade at a discount to CSU on a gaap basis?

Subject Re: Organic Growth
Entry 02/16/2026 02:40 PM
Member Djokovic1

mrbing,

It's a good point you bring up but your framework is wrong. You can think of the M&A spend as capex but you have to distinguishing between growth and maintenance capex. Any M&A spend can be thought of as growth capex (however I just prefer thinking of it as M&A spend but the math will lead you to the same end point).

The important point is that the FCF or owners earnings would take away the maintenance capex (not growth capex or the M&A spend). Same if a company deployed all its FCF into share buybacks, that doesn't mean its FCF is 0.

Subject Re: Re: Organic Growth
Entry 02/16/2026 03:19 PM
Member Djokovic1

I would also add that just looking at organic growth (and comparing to other SaaS companies) is not the way to value Constellation software. Majority of their historic growth has been inorganic with a small minority of ~3% from organic growth.

The question is not whether CSU should be valued at the same multiple as other SaaS companies growing LSD organically, that's not apples to apples. The questions are i) to what extent can AI disrupt VMS and ii) going forward what level of inorganic growth will CSU achieve in addition to its organic growth

Subject Modeling
Entry 02/16/2026 08:58 PM
Member jim211

I don't think I have ever done another company this way but when I modeled Constellation clearly a big locus of value creation is, or at least has been, the acquisitions. So my major assumptions on that were simply % of free cash flow deployed and price/sales ratio paid. Law of large numbers makes me fade these in the model. But the cash flows I discount are only those not deployed to acquisitions, which just means a very big terminal value. So yes, if one is "counting" the value creation from the acquisitions, that is a cash outflow.

Subject Re: Modeling
Entry 02/17/2026 08:48 AM
Member ahnuld

was gonna say, im surprised this is being debated as its quite simple to model. Just build in acquisitions to your model, assume a sales multiple acquisition price, deduct the cash flow they spend on M&A, and then use a terminal multiple. I did it in my super basic Transcontinental model I posted for my writeup. Screenshot just to show you how id think about it.

EBITDA organic	207	209	209	215	220	225
Acquired			17	17	17	17
Total EBITDA			225	231	237	242
Capex		55	60	60	60	60
Cash EBIT		154	149	155	160	165
Interest		22	22	19	16	12
EBT		132	127	136	145	154
Cash Tax		36	30	32	34	41
FCF		96	97	104	111	112
FCF after acquisitions			47	54	61	62
EBITDA margins	17.7%	18.2%	19.2%	18.8%	18.5%	18.2%
Debt		360	313	260	199	137
Stub Price Target	5.67					
Acquisition adjusted FCF						
FCFE		38.4%	9.8%	11.4%	12.8%	13.1%
FCFF			8.7%	9.9%	11.3%	12.1%
Organic decline			-5%	-3%	-3%	-3%
Sales/EBITDA decline flowthrough			0%	30%	30%	30%
Spend	3x		50	50	50	50
Acquired EBITDA			16.7	16.7	16.7	16.7

Subject	Sabre Corporation (SABR) & Constellation Software
Entry	03/04/2026 08:25 AM
Member	coffee1029

"Between April 2025 and November 2025, Constellation accumulated a 9.7% economic position in Sabre, comprising 4.7% beneficial ownership of common stock and a further 5% via derivative instruments, and privately informed Sabre of its ownership stake for the first time in early January 2026"

<https://investors.sabre.com/news-releases/news-release-details/sabre-corporation-adopts-limited-duration-shareholder-rights>

1) Without picking at scabs, any former or current Sabre longs here consider Constellation's investment at a fraction of VIC write-up prices sensible? Or shorts who cite this as more evidence that Constellation is extinction-bound?

2) MarAzul - any thoughts on how this plays out?

3) FWIW Mark Miller has struck me over the years as a bit more driven, and more obviously focused on organic growth opportunities than some other senior managers at Constellation. I found it interesting therefore that on his succession last year he expected large inorganic opportunities to be taking up a lot of his time. CSI's data on tiny acquisitions has been a real edge, and I would expect that to continue, with adaptations in the AI era. Their large acquisition dataset is clearly much slimmer, so they need to find different edges. I guess this one might end up being the first real time case study under new leadership.

Subject Re: Sabre Corporation (SABR) & Constellation Software
Entry 03/04/2026 08:34 AM
Member mrbing

SABR has basically been distressed since covid and pre-covid were losing share pretty consistently to AMS. Not that long ago they sold their best asset, the Hospitality business, to re-pay expensive debt they raised to avoid bankruptcy a couple years ago. They have \$4.5 billion debt and are levered 7x. Most of their debt was trading at 20% YTM's just this past Friday. Seems silly to me, but I'm uninvolved one way or another.

	Rate	Maturity	December 31,	
			2025	2024
Senior Secured Credit Facilities:				
2021 Term Loan B-1	S ⁽¹⁾ + 3.50%	December 2027	\$ —	\$ 314,860
2021 Term Loan B-2	S ⁽¹⁾ + 3.50%	December 2027	—	366,280
2022 Term Loan B-1	S ⁽¹⁾ + 4.25%	June 2028	—	382,147
2022 Term Loan B-2	S ⁽¹⁾ + 5.00%	June 2028	—	414,482
2024 Term Loan B-1	S ⁽¹⁾ + 6.00%	November 2029	397,100	790,000
2024 Term Loan B-2	S ⁽¹⁾ + 6.00%	November 2029	74,250	75,000
2025 Term Loan B-1	S ⁽¹⁾ + 6.25%	July 2029	288,240	—
2025 Term Loan B-2	S ⁽¹⁾ + 6.25%	July 2029	86,760	—
2026 Term Loan	RR ⁽²⁾ + 1.75% ⁽³⁾	December 2028	—	871,811
Securitization facility:				
AR Facility	S ⁽¹⁾ + 4.00% ⁽⁴⁾	March 2027	82,000	82,200
FILE Facility	S ⁽¹⁾ + 8.00%	March 2027	120,000	120,000
9.25% senior secured notes due 2025	9.25%	April 2025	—	10,416
4.00% senior exchangeable notes due 2025	4.00%	April 2025	—	183,220
7.375% senior secured notes due 2025	7.375%	September 2025	—	23,393
7.32% senior exchangeable notes due 2026	7.32%	August 2026	150,000	150,000
8.625% senior secured notes due 2027	8.625%	June 2027	91,607	656,783
11.25% senior secured notes due 2027	11.25%	December 2027	1,558	45,814
10.75% senior secured notes due 2029	10.75%	November 2029	445,715	824,714
11.125% senior secured notes due 2030	11.125%	July 2030	1,325,000	—
10.75% senior secured notes due 2030	10.75%	March 2030	469,802	—
11.125% senior secured notes due 2029	11.125%	June 2029	1,080,000	—
Face value of total debt outstanding			4,532,032	5,220,920
Less current portion of debt outstanding			(247,665)	(230,704)
Face value of long-term debt outstanding			\$ 4,284,367	\$ 4,990,216

Subject Re: Sabre Corporation (SABR) & Constellation Software
Entry 03/04/2026 11:50 PM
Member MarAzul

No idea what is the end game here. Seems like some sort of restructuring is the likely outcome here. Not clear what is the plan, but we will soon find out I guess.

Subject FCF calculation
Entry 04/11/2026 12:48 PM
Member finn520

What FCF number do you get for 2025? Is it different than the company's reported \$1,683 million FCF available to shareholders in the press release (they call it FCFA2S)? Why does the company subtract the IRGA/TSS Membership liability charge in their FCFA2S number? Isn't that basically contingent consideration for past acquisition and should be thought of as some future liability rather than an annual subtraction from FCF? They don't subtract other acquisitions from the FCFA2S number. Not sure that I am thinking about this right.

Thanks.

Subject Re: FCF calculation
Entry 04/11/2026 01:57 PM
Member churchill

if it helps, CSU's FCFA2S number removes the TOI and LMN FCF it isn't actually entitled to.

However, focusing on the % of FCFA2S deployed in acquisitions by CSU is wrong, as CSU can only deploy CSU's FCF and not its proportional share of TOI and LMN's FCF

On the other hand, the IRGA liability which is deducted from CSU's FCFA2S is cash they do have in the bank to deploy. But I don't see it as a liability. It is not a contingent consideration, it is a put option from Joday which, if exercised, would give CSU an asset.

Subject Re: Re: FCF calculation
Entry 04/11/2026 03:49 PM

Member
Churchill,

finn520

I am aware that CSU's FCFA2s number removes the TOI and LMN it isn't entitled to. I think that is the proper treatment.

I am not focused on the # of FCFA2S deployed in acquisitions by CSU for the purpose of understanding their FCF number. It's important as it relates to analyzing the company and future growth, but that's a separate issue.

I don't understand why they deduct the TSS/IRGA liability revaluation charge in their FCFA2S number. It doesn't make sense to me as I would expect the FCFA2S number to represent the FCF produced by the company in any given year that is available to be allocated, via acquisitions or dividends or repurchase or whatever.

I guess we can disagree over what to call the TSS/IRGA arrangement. I refer to it as a liability because it is literally a liability on the CSU balance sheet - at year end 2025 \$775 million current + \$458 million LT = \$1,233 million total. I get that they receive an asset in return for that payment, that asset being the remaining minority interest in Topcius. Do you know the terms of that arrangement and how is likely to work in practice? I see a reference on Reddit that it's at 3x maintenance revenue, which is way below Topicus's current multiple. If that's the case, it ties back to my original question of why they are deducting this yearly change from the FCFA2S number.

Subject Re: Re: Re: FCF calculation
Entry 04/11/2026 05:41 PM
Member churchill

I personally add it back. I believe they subtract it to highlight uncommitted FCF. As far as I know, the formula isn't public, but you can approximate it by comparing how the liability has evolved relative to Topicus' maintenance revenue and 3x seems to be directionally right.

Subject Re: FCF calculation
Entry 04/12/2026 09:33 AM
Member Trajan

To me this is a really conservative position adopted by the management but it is also rational. I mean, should the minority shareholders exercise the option, the payment would actually absorb the cash flow or dilute the shareholders.

They are essentially amortizing the future cost of the buyout in real-time as the business grows.

If CSU waited until the year the option was actually exercised to recognize the cash outflow, it would result in a massive artificial plunge in FCFA2S in that specific year, while heavily overstating the true unencumbered cash generation in all the preceding years.

Best

Subject Author Exit Recommendation
Entry 06/05/2026 08:58 PM
Member MarAzul

The author has recommended exiting the position

Subject Re: Author Exit Recommendation
Entry 06/05/2026 09:23 PM
Member VC2020

Curious if you could elaborate as to why the exit?

Subject Re: Re: Author Exit Recommendation
Entry 06/06/2026 03:23 PM
Member MarAzul

Hi VC,

It is not a negative view on the company. More a personal decision as I did not feel comfortable holding through the various swings; don't have confidence on how all these technological changes will affect the business long term. Probably no one knows as they own a ton of different businesses in various sectors, etc.

I still think they have an excellent culture, aligned management, close to zero net debt at holdco, no sbc, etc. Also they have been deploying a ton of \$, and the environment seems fertile for investing capital. So there are many positives. There is a home run scenario where AI helps them build capabilities that are distributed to their customers and makes the business larger, stickier, etc. who knows.

The advances on AI have been surprising over the past few months and just don't know how things will turn out. Prefer holding cash or investing my money elsewhere.

Subject Re: Re: Re: Author Exit Recommendation
Entry 06/06/2026 03:33 PM
Member VC2020

Certainly agree with essentially everything you said. I own this for many of the same reasons, and I also share many of the same concerns (and have kept it on the smaller side due to the latter). The advancements in AI have been quite astounding (and concerning).